# Product Requirements Document

## Sales Call Intelligence Platform

**Document version:** 1.0
**Product stage:** Internal Production
**Deployment model:** Single-company, private deployment
**Primary users:** Sales managers, sales representatives, system administrators
**Status:** Proposed

---

## 1. Executive Summary

The Sales Call Intelligence Platform automatically retrieves recorded sales calls from Microsoft Dynamics, converts Persian audio into text using Farsava, analyzes the transcript using AI, stores the results, and presents actionable insights through an internal management dashboard.

The platform will help the company:

- Review sales calls without manually listening to every recording.
- Identify high-potential leads.
- Evaluate sales representative performance.
- Detect common objections, weaknesses, and coaching opportunities.
- Improve visibility into sales quality and customer sentiment.
- Create a consistent, searchable record of call outcomes.

This release is intended exclusively for internal company use. It must be reliable, secure, observable, and recoverable, but it does not require external customer onboarding, billing, multi-tenancy, public APIs, or commercial-grade tenant isolation.

---

## 2. Problem Statement

Sales managers currently have limited visibility into the quality and outcome of sales conversations. Reviewing calls manually is time-consuming and inconsistent, making it difficult to:

- Identify valuable leads quickly.
- Coach sales representatives using objective evidence.
- Detect repeated customer objections.
- Compare performance across teams and employees.
- Monitor whether required sales practices are followed.
- Track the relationship between call quality and deal outcomes.

The company needs an automated internal platform that processes recorded calls and converts them into structured, searchable, and reviewable sales intelligence.

## 3. Product Goals

### 3.1 Primary Goals

1. Automatically process eligible sales calls received from Microsoft Dynamics.
2. Produce accurate Persian-language transcripts through Farsava.
3. Generate structured AI analysis for every successfully transcribed call.
4. Store all processing states and results reliably.
5. Provide managers with a secure dashboard for reviewing calls and team performance.
6. Ensure that temporary failures do not cause calls to be permanently lost.
7. Provide sufficient monitoring and auditability for internal production operations.

### 3.2 Success Definition

The internal production release will be considered successful when:

- At least 99.5% of valid calls are eventually processed, excluding confirmed provider outages or invalid source data.
- No acknowledged webhook event is silently lost.
- Duplicate events do not create duplicate final analyses.
- Managers can locate and review processed calls without technical assistance.
- Failed calls are visible and can be retried by an administrator.
- Operational alerts identify outages before business users report them.
- Access to recordings, transcripts, and analyses is restricted to authorized employees.

---

## 4. Non-Goals

The first internal production release will not include:

- External customer access.
- Multi-tenant architecture.
- Subscription billing.
- Public developer APIs.
- Marketplace integrations.
- Customer-managed configuration.
- Automated employee disciplinary decisions.
- Fully autonomous CRM updates based on AI conclusions.
- Real-time call assistance during an active call.
- Support for multiple speech-to-text providers.
- Support for languages other than Persian unless separately approved.
- Native mobile applications.
- Advanced predictive sales forecasting.
- Automatic modification of Dynamics opportunities without human approval.

---

# 5. Users and Roles

## 5.1 Sales Manager

A sales manager needs to:

- Review calls for their team.
- View lead scores and call quality scores.
- identify strengths, weaknesses, objections, and recommended coaching actions.
- Filter calls by employee, date, status, and score.
- Open the transcript and call analysis.
- Listen to the original recording when authorized.
- Mark an analysis as reviewed.
- Add internal notes.

## 5.2 Sales Representative

A sales representative may be permitted to:

- View their own calls.
- Review coaching feedback.
- View strengths and improvement areas.
- Add comments or context.
- Dispute an incorrect analysis.

Access to comparative team-level data will depend on company policy.

## 5.3 System Administrator

A system administrator needs to:

- Monitor service health.
- View failed and delayed jobs.
- Retry eligible failed jobs.
- Disable or re-enable integrations.
- Manage access roles.
- Review audit logs.
- Configure non-secret application settings.
- Confirm the status of database migrations and external providers.

## 5.4 Executive Viewer

An executive viewer may access:

- Aggregated performance statistics.
- Team-level trends.
- High-level call processing metrics.
- Lead and quality score distributions.

This role should not automatically receive access to full recordings or transcripts.

---

## 6. Assumptions

This PRD assumes:

- Microsoft Dynamics contains a stable identifier for each call.
- Each eligible call includes or references a retrievable audio recording.
- Farsava supports the required audio formats and Persian transcription.
- The company has legal authorization to record, process, and analyze the relevant calls.
- The company will define which roles may access audio, transcripts, and employee evaluations.
- The application will initially serve one company and one primary Dynamics environment.
- The platform will be deployed in a private company-controlled environment or approved cloud account.
- AI-generated results are advisory and must not be treated as unquestionable facts.

---

## 7. End-to-End User Journey

### 7.1 Automated Processing Journey

1. A sales call is completed and recorded.
2. Microsoft Dynamics sends a webhook event or the synchronization service discovers the call.
3. The backend authenticates and validates the event.
4. The event and an outbox record are stored in one database transaction.
5. The backend returns an acknowledgment without downloading or processing the audio.
6. A background dispatcher places the call-processing job into the queue.
7. A worker retrieves the call metadata and downloads the recording.
8. The worker validates the audio type, size, checksum, and source.
9. The recording is stored in approved object storage.
10. The worker submits the audio to Farsava.
11. The returned transcript is validated and stored.
12. The transcript is sent to the AI analysis workflow.
13. The AI produces a validated structured response.
14. The analysis and final processing status are stored.
15. The call becomes visible in the dashboard.
16. Temporary data is deleted according to the retention policy.

### 7.2 Manager Review Journey

1. The manager signs in through the company authentication system.
2. The dashboard displays recent call activity and summary statistics.
3. The manager filters calls by employee, team, date, score, or status.
4. The manager selects a call.

5. The detail page displays:

6. Call metadata.

7. Processing status.
8. Transcript.
9. Lead score.
10. Sales quality score.
11. Strengths.
12. Weaknesses.
13. Objections.
14. Suggested coaching actions.
15. AI confidence or warning indicators.
16. The manager optionally plays the recording.
17. The manager marks the call as reviewed and adds a note.

### 7.3 Failure Recovery Journey

1. A provider request or internal processing step fails.
2. The system classifies the error as temporary or permanent.
3. Temporary errors are retried using exponential backoff and jitter.
4. Permanent errors or exhausted retries are placed in a failed-job state.
5. An alert is created when failure thresholds are exceeded.
6. An administrator reviews the failure reason.
7. The administrator may retry the call after correcting the cause.
8. Processing resumes from the last completed stage whenever possible.

---

# 8. Functional Requirements

# 8.1 Microsoft Dynamics Integration

The system must:

- Authenticate securely with Microsoft Dynamics.
- Accept webhook events through a protected endpoint.
- Support scheduled synchronization as a recovery mechanism.
- Validate required call identifiers and recording references.
- Reject malformed or unauthorized events.
- Store the original external event identifier.
- Detect duplicate webhook deliveries.
- Avoid processing the same call version more than once.
- Apply request and connection timeouts.
- Respect provider throttling and `Retry-After` responses.
- Record provider request identifiers where available.
- Support configurable synchronization windows.
- Avoid downloading audio within the webhook request lifecycle.

### Acceptance Criteria

- A valid webhook receives a successful acknowledgment after durable event storage.
- Duplicate webhook events do not create duplicate call analyses.
- Invalid authentication returns an appropriate authorization error.
- Malformed payloads are rejected and safely logged.
- A missed webhook can be recovered through scheduled synchronization.

---

## 8.2 Queue and Background Processing

The system must use asynchronous background processing for all network-intensive and AI-related work.

The queue system must support:

- Unique job identifiers.
- Configurable retry limits.
- Exponential backoff.
- Randomized retry jitter.
- Per-stage timeouts.
- Job status tracking.
- Dead-letter or permanent-failure handling.
- Detection of stuck jobs.
- Controlled worker concurrency.
- Graceful shutdown.
- Safe job recovery after worker restart.
- Queue-depth monitoring.
- Manual retry by authorized administrators.

The system must not assume that a queued task executes only once. Every processing stage must be idempotent.

---

## 8.3 Audio Retrieval and Storage

The system must:

- Stream audio downloads rather than loading entire files into memory.
- Apply a maximum supported file size.
- Validate the content type and file signature.
- Generate an internal object key instead of trusting source filenames.
- Calculate and store a checksum.
- Encrypt recordings in transit and at rest.
- Prevent path traversal and unsafe filenames.
- Record storage location, size, checksum, and retention date.
- Delete temporary local files after processing.

- Support retry without downloading the same valid recording unnecessarily.
- Prevent unauthorized direct access to stored recordings.
- Use time-limited signed access when recordings are played from the dashboard.

Shared local Docker volumes may be used for development, but internal production must use durable storage accessible by all workers.

---

## 8.4 Farsava Speech-to-Text Integration

The system must:

- Submit supported audio formats to Farsava.
- Use configurable connection and response timeouts.
- Handle asynchronous provider jobs if required.
- Store the provider request identifier.
- Handle rate limiting.
- Retry temporary failures.
- Avoid retrying invalid credentials or invalid audio indefinitely.
- Store the original transcription response for troubleshooting when permitted.
- Validate that a usable transcript was returned.
- Mark empty or severely incomplete transcripts for review.
- Expose transcription failure reasons to administrators.
- Record transcription duration and provider latency.

### Transcript Requirements

Each transcript should contain, when supported:

- Full text.
- Speaker separation.
- Timestamps.
- Detected language.
- Provider confidence.
- Audio duration.

The product must continue operating when optional transcript fields are unavailable.

---

## 8.5 AI Analysis

The AI workflow must produce a versioned structured result validated by Pydantic or an equivalent schema.

The minimum analysis output must include:

- Lead score from 0 to 100.
- Overall call quality score from 0 to 100.

- Customer interest level.
- Call summary.
- Customer needs.
- Customer objections.
- Positive buying signals.
- Sales representative strengths.
- Sales representative weaknesses.
- Recommended coaching actions.
- Recommended next action.
- Risk indicators.
- Evidence excerpts or transcript references supporting major conclusions.
- Analysis model version.
- Prompt or workflow version.
- Processing timestamp.

The system must:

- Use asynchronous model invocation.
- Apply a maximum execution time.
- Limit transcript and prompt size.
- Avoid including unnecessary personal or sensitive data.
- Validate every AI response before storage.
- Retry schema-format failures a limited number of times.
- Mark unverifiable or low-confidence results.
- Treat transcript content as untrusted input.
- Defend against prompt-injection instructions contained inside transcripts.
- Prevent the model from triggering tools or external actions unless explicitly designed and approved.
- Store enough version information to reproduce or compare analyses.
- Allow re-analysis when the workflow version changes.
- Clearly label analysis as AI-generated.

### AI Safety Requirement

AI scores must not automatically:

- Change employee compensation.
- Trigger disciplinary action.
- Modify a CRM deal stage.
- Contact a customer.
- Reject a lead.
- Make legal or compliance determinations.

Such decisions require human review.

---

## 8.6 Database and Processing State

The system must use PostgreSQL as the system of record.

Each call must have a stable unique identifier and a state such as:

- `RECEIVED`
- `QUEUED`
- `DOWNLOADING_AUDIO`
- `AUDIO_READY`
- `TRANSCRIBING`
- `TRANSCRIBED`
- `ANALYZING`
- `COMPLETED`
- `RETRY_SCHEDULED`
- `FAILED_PERMANENT`
- `CANCELLED`

The system must store:

- External call ID.
- External event ID.
- Call metadata.
- Owner or sales representative.
- Processing status.
- Attempt count.
- Last successful stage.
- Last error code.
- Sanitized error message.
- Next retry time.
- Provider request IDs.
- Audio metadata.
- Transcript.
- AI analysis.
- Model and workflow version.
- Creation and update timestamps.
- Review status.
- Manager notes.
- Audit metadata.

Database writes must:

- Use transactions.
- Enforce unique constraints.
- Prevent duplicate final records.
- Support safe upsert behavior.
- Avoid partially completed final results.
- Use migration scripts for all schema changes.
- Preserve previous schema compatibility during rolling deployment where required.

---

## 8.7 Dashboard

The dashboard must be responsive and usable on standard company desktop and laptop screens.

### Dashboard Overview

The landing page must display:

- Calls received today.
- Calls completed today.
- Calls currently processing.
- Failed calls.
- Average lead score.
- Average quality score.
- Calls requiring review.
- Recent calls.
- Processing trend over time.

### Call List

Users must be able to filter by:

- Date range.
- Sales representative.
- Team.
- Processing status.
- Review status.
- Lead score range.
- Quality score range.
- Customer or account.
- Call identifier.

The list must support:

- Pagination.
- Server-side filtering.
- Sortable columns.
- Loading states.
- Empty states.
- Error states.
- Refresh.
- Stable URLs for filters where practical.

### Call Detail

The call detail view must display:

- Call metadata.

- Audio player, when authorized.
- Transcript.
- AI summary.
- Lead score.
- Quality score.
- Needs and objections.
- Strengths and weaknesses.
- Recommended actions.
- Supporting evidence.
- Processing timeline.
- Analysis version.
- Review status.
- Internal notes.
- Dispute or feedback control.

### Dashboard Safety

The frontend must not expose:

- Provider credentials.
- Database credentials.
- Internal stack traces.
- Raw authorization tokens.
- Recordings through permanent public links.
- Information belonging to teams the user cannot access.

---

## 8.8 Authentication and Authorization

Internal production must integrate with the company identity provider whenever possible.

Preferred authentication:

- Microsoft Entra ID or the company-approved SSO provider.
- Multi-factor authentication enforced by organizational policy.
- Short-lived sessions.
- Secure, HTTP-only cookies where applicable.
- Logout and session expiration.

The system must implement role-based access control.

Authorization must be enforced by the backend, not only hidden in the user interface.

Minimum permissions:

| Permission | Admin | Manager | Representative | Executive |
|---|---|---|---|---|
| View system health | Yes | No | No | No |
| Retry failed jobs | Yes | No | No | No |
| Manage access | Yes | No | No | No |
| View team calls | Yes | Yes | No | Optional |
| View own calls | Yes | Yes | Yes | No |
| Play recordings | Policy-based | Policy-based | Policy-based | No by default |
| Add review notes | Yes | Yes | Optional | No |
| View aggregate metrics | Yes | Yes | Limited | Yes |

---

## 8.9 Administration

The system must provide an internal administrative area or equivalent protected operational interface for:

- Viewing failed jobs.
- Viewing delayed and stuck jobs.
- Retrying eligible jobs.
- Viewing provider status.
- Viewing recent processing errors.
- Inspecting pipeline stage history.
- Confirming application version.
- Confirming database migration version.
- Enabling or disabling scheduled synchronization.
- Viewing queue depth.
- Viewing recent configuration changes.

Secrets must not be editable or displayed through the application interface.

---

## 8.10 Notifications and Alerts

Operational alerts must be sent through an approved internal channel such as email, Microsoft Teams, or the company monitoring platform.

Alerts must cover:

- Repeated webhook authentication failures.
- Queue backlog above threshold.
- Oldest queued job above threshold.
- Worker unavailable.

- Redis unavailable.
- PostgreSQL unavailable.
- Farsava failure rate above threshold.
- Dynamics failure rate above threshold.
- Processing success rate below threshold.
- Disk or object storage capacity risk.
- Database backup failure.
- Dead-letter job growth.
- Unusual authentication or authorization failures.

Alerts must include a traceable identifier and must not contain full transcripts, recordings, credentials, or sensitive customer information.

---

# 9. Non-Functional Requirements

## 9.1 Performance

Initial performance targets:

- Webhook acknowledgment p95: less than 500 milliseconds after durable event storage.
- Dashboard summary API p95: less than 500 milliseconds under expected internal load.
- Call list API p95: less than 750 milliseconds for standard filtered queries.
- Call detail API p95: less than 750 milliseconds, excluding audio delivery.
- Dashboard initial usable render: less than 3 seconds on the company network.
- No full audio file may be loaded into API or worker memory.
- Worker concurrency must be configurable.
- Database queries used by the dashboard must use appropriate indexes.
- Large transcript responses must support compression and controlled payload sizes.

Processing time for transcription and AI analysis depends on external providers and audio length. The system must display progress rather than presenting provider latency as an application failure.

---

## 9.2 Reliability

The platform must:

- Avoid losing events after acknowledging them.
- Recover from API, worker, Redis, or database restarts.
- Resume from the last completed processing stage.
- Prevent duplicate final analyses.
- Support graceful worker shutdown.
- Make temporary external-provider outages recoverable.
- Isolate permanent failures from healthy jobs.
- Preserve the original call record when re-analysis fails.

- Use health checks suitable for orchestration.
- Provide documented recovery procedures.

### Availability Target

For the initial internal release:

- Monthly application availability target: 99.5%.
- Planned maintenance may be excluded when communicated internally.
- External provider outages must be measured separately.
- Dashboard read access should remain available when AI providers are unavailable.
- Ingestion should continue or safely buffer when downstream providers are unavailable.

---

## 9.3 Health Endpoints

The backend must provide separate endpoints for:

- Liveness: confirms that the application process is running.
- Readiness: confirms that the service can accept traffic.
- Startup: confirms that initialization and required migrations are complete.

Liveness must not fail solely because an external provider is temporarily unavailable.

Health endpoints must not disclose credentials, internal connection strings, or detailed infrastructure information.

---

## 9.4 Security

The application must follow these baseline controls:

- TLS for all network communication.
- Encryption at rest for recordings, transcripts, backups, and databases.
- Secrets stored in an approved secrets manager or protected deployment environment.
- No real secrets committed to source control.
- Secret rotation procedure.
- Least-privilege service accounts.
- Backend authorization on every protected resource.
- Rate limiting on sensitive endpoints.
- Request body size limits.
- Strict schema validation.
- Secure response headers.
- Content Security Policy for the dashboard.
- Protection against cross-site scripting and cross-site request forgery where applicable.
- Safe CORS configuration using explicit origins.

- Dependency and container image scanning.
- Non-root containers.
- Removal of unnecessary Linux capabilities.
- Read-only container filesystems where practical.
- Audit logging for privileged actions.
- Protection against log injection.
- Redaction of credentials, tokens, and sensitive personal data from logs.

Webhook security must include:

- Strong shared-secret or signed-request verification.
- Constant-time secret comparison.
- Timestamp or replay protection where supported.
- Idempotency controls.
- Rate limits.
- Secret rotation without extended downtime.

---

## 9.5 Privacy and Data Governance

Before launch, the company must approve:

- Which calls may be recorded and analyzed.
- Required caller notification or consent.
- Who may access recordings.
- Who may access transcripts.
- Who may access employee performance analysis.
- Data retention periods.
- Deletion procedures.
- Legal hold procedures.
- Employee dispute and correction procedures.
- Whether sensitive categories must be redacted.

The system must support separate retention settings for:

- Raw recordings.
- Transcripts.
- AI analysis.
- Operational logs.
- Audit logs.
- Provider responses.
- Backups.

Deletion must cover primary storage and follow documented backup-expiration behavior.

---

## 9.6 Observability

The platform must produce:

- Structured application logs.
- Distributed traces.
- Operational metrics.
- Business processing metrics.
- Alertable health indicators.

Every processing flow must include:

- `call_id`
- `event_id`
- `job_id`
- `trace_id`
- External provider request IDs where available

Required metrics include:

- Webhooks received.
- Webhooks rejected.
- Duplicate events.
- Queue depth.
- Oldest queued-job age.
- Active workers.
- Processing duration by stage.
- Retry counts.
- Permanent failures.
- Provider latency.
- Provider error rate.
- Database pool utilization.
- Slow queries.
- Completed calls.
- Calls awaiting review.
- Processing success ratio.

Logs must not include complete recordings, full transcripts, API keys, access tokens, passwords, or sensitive headers.

---

## 9.7 Backup and Disaster Recovery

The production environment must include:

- Automated PostgreSQL backups.
- Point-in-time recovery where supported.

- Encrypted backups.
- Defined backup retention.
- Object-storage durability and lifecycle rules.
- Redis persistence or an architecture that does not treat Redis as the only durable record.
- Documented restoration procedures.
- Scheduled restore testing.
- Configuration and deployment backups.

Initial recovery objectives:

- Recovery Point Objective: no more than 15 minutes for database data.
- Recovery Time Objective: no more than 4 hours for the internal service.
- No acknowledged ingestion event should depend solely on Redis for durability.

---

## 9.8 Scalability

The initial platform must support horizontal worker scaling without duplicate processing.

The architecture must allow:

- Multiple API instances.
- Multiple worker instances.
- Shared object storage.
- Central PostgreSQL.
- Highly available or managed Redis.
- Stateless API containers.
- Configurable concurrency.
- Queue-based backpressure.

Capacity testing must use at least twice the expected peak call-ingestion rate before launch.

---

## 9.9 Maintainability

The project must include:

- Type checking for Python and TypeScript.
- Linting.
- Automated formatting.
- Unit tests.
- Integration tests.
- End-to-end tests.
- Migration tests.
- Documented configuration.
- Runbooks.
- API documentation.

- Versioned AI schemas.
- Versioned prompts or workflow definitions.
- Clear separation between development, staging, and production.

All production dependencies and container base images must be pinned to controlled versions.

---

# 10. Testing Requirements

## 10.1 Unit Tests

Unit tests must cover:

- Configuration validation.
- Authentication helpers.
- Webhook validation.
- Pydantic schemas.
- Status transitions.
- Retry classification.
- Idempotency logic.
- Score validation.
- Authorization decisions.
- AI response parsing.

## 10.2 Integration Tests

Integration tests must cover:

- FastAPI and PostgreSQL.
- FastAPI and Redis.
- Worker job execution.
- Database upsert behavior.
- Transactional outbox behavior.
- Duplicate event processing.
- Migration from the previous database version.
- Object storage upload and deletion.
- Mock Dynamics responses.
- Mock Farsava responses.
- Mock AI responses.

## 10.3 End-to-End Tests

The staging environment must verify:

1. A call enters from Dynamics.
2. The webhook is acknowledged.

3. The job is queued.
4. Audio is retrieved.
5. Transcription completes.
6. AI analysis completes.
7. Results appear in the dashboard.
8. Access control is enforced.
9. Failed jobs can be retried.
10. Duplicate events do not duplicate results.

At least one pre-launch test must use approved real sandbox or production-like credentials for each external provider.

## 10.4 Failure Tests

The platform must be tested under:

- Dynamics timeout.
- Farsava timeout.
- AI timeout.
- HTTP 429.
- HTTP 500.
- Invalid audio.
- Empty transcript.
- Invalid AI JSON.
- Redis restart.
- Worker termination during processing.
- Database restart.
- Duplicate webhook delivery.
- Object-storage failure.
- Deployment during queued work.

## 10.5 Load Tests

Load testing must confirm:

- Webhook latency remains within target.
- Queue backlog remains recoverable.
- Worker scaling behaves predictably.
- PostgreSQL connection limits are not exceeded.
- Dashboard performance remains acceptable.
- No events are silently lost.
- Rate limits and backpressure work as designed.

## 10.6 Security Tests

Before launch:

- Dependency scan.
- Container scan.
- Secret scan.
- Authentication and authorization testing.
- Webhook replay testing.
- Rate-limit testing.
- File-upload and path-traversal testing.
- Cross-site scripting testing.
- CORS validation.
- Security-header validation.
- Review of sensitive data in logs.

---

# 11. Deployment Requirements

The platform must have separate:

- Development environment.
- Staging environment.
- Production environment.

Production deployment must include:

- Automated build.
- Automated tests.
- Database migration checks.
- Container image scanning.
- Versioned image publishing.
- Controlled deployment.
- Health verification.
- Rollback procedure.
- Deployment audit trail.
- Configuration validation.
- Post-deployment smoke tests.

Production must not deploy directly from a developer workstation.

Database migrations must be reviewed for:

- Backward compatibility.
- Lock duration.
- Rollback or forward-recovery strategy.
- Impact on existing workers and API instances.

## 12. Internal Rollout Plan

### Phase 1: Technical Validation

Participants:

- Engineering.
- IT operations.
- Security.
- One sales operations representative.

Goals:

- Confirm integrations.
- Validate monitoring.
- Test failure recovery.
- Confirm retention and access rules.

### Phase 2: Limited Pilot

Participants:

- One sales team.
- One manager.
- A limited number of representatives.

Goals:

- Validate analysis usefulness.
- Measure transcription quality.
- Identify incorrect AI assumptions.
- Confirm dashboard usability.
- Gather dispute and feedback patterns.

### Phase 3: Department Rollout

Participants:

- All approved sales teams.

Requirements:

- Pilot blockers resolved.
- Support process active.
- Documentation published.
- Monitoring and alert ownership assigned.
- Data governance approval completed.

### Phase 4: Optimization

Potential enhancements:

- Better coaching analytics.
- Trend analysis.
- CRM write-back with approval.
- Improved speaker separation.
- Custom score calibration.
- Additional dashboards.
- Alternative transcription provider fallback.

---

## 13. Product Metrics

### Operational Metrics

- Percentage of valid calls successfully completed.
- Median and p95 processing time.
- Queue backlog.
- Permanent failure rate.
- Retry rate.
- Provider error rate.
- Duplicate-event rate.
- Dashboard API latency.
- Service availability.

### Adoption Metrics

- Weekly active managers.
- Percentage of completed calls reviewed.
- Average time from processing completion to review.
- Number of manager notes.
- Number of representative feedback submissions.

### Quality Metrics

- Transcript correction rate.
- Manager agreement with lead score.
- Manager agreement with quality score.
- Percentage of analyses marked inaccurate.
- Percentage of AI claims with supporting transcript evidence.
- False positive rate for major risk indicators.

### Business Metrics

Business impact should be measured cautiously and may include:

- Reduction in manual call-review time.
- Increase in coaching frequency.
- Faster identification of high-potential leads.
- Improvement in follow-up completion.
- Changes in conversion rate after sufficient data is collected.

The product must not claim that AI caused business improvement without an appropriate evaluation methodology.

---

## 14. Risks and Mitigations

### AI Inaccuracy

**Risk:** Scores or coaching feedback may be incorrect.

**Mitigation:** Human review, evidence references, confidence indicators, feedback controls, workflow versioning, and no autonomous employment decisions.

### Transcription Errors

**Risk:** Persian accents, noise, or overlapping speakers may reduce transcript quality.

**Mitigation:** Store confidence data, expose audio playback, identify low-quality transcripts, and support reprocessing.

### Provider Outage

**Risk:** Dynamics, Farsava, or AI services may be temporarily unavailable.

**Mitigation:** Durable ingestion, retries, backoff, provider monitoring, dead-letter handling, and resumable stages.

### Duplicate Processing

**Risk:** Repeated webhooks or worker retries may create duplicate costs or records.

**Mitigation:** Database uniqueness, idempotent stages, unique job IDs, and transactional outbox.

### Sensitive Data Exposure

**Risk:** Recordings and transcripts may contain personal or confidential information.

**Mitigation:** SSO, backend authorization, encryption, retention limits, access auditing, secure links, and log redaction.

### Employee Trust

**Risk:** Employees may consider the tool unfair or overly intrusive.

**Mitigation:** Transparent policies, access to personal results, correction and dispute process, human review, and prohibition of fully automated disciplinary action.

### Cost Growth

**Risk:** Reprocessing or uncontrolled AI usage may increase provider costs.

**Mitigation:** Stage-level caching, idempotency, token limits, cost metrics, usage alerts, and administrator-controlled re-analysis.

---

## 15. Launch Blockers

The application must not enter internal production until all of the following are complete:

- Company SSO or approved secure authentication.
- Backend-enforced RBAC.
- Durable webhook ingestion.
- Transactional outbox or equivalent no-loss mechanism.
- Idempotent processing.
- Durable shared object storage.
- Retry, timeout, backoff, and permanent-failure handling.
- Production PostgreSQL backups.
- Restore test completed successfully.
- Redis persistence and recovery strategy.
- Structured logs, metrics, and alerts.
- Audit logs for privileged actions.
- Production secret management.
- Container hardening.
- Staging end-to-end test with external integrations.
- Load test at twice expected peak.
- Worker crash and restart test.
- Database migration and rollback procedure.
- Security review.
- Data-retention policy.
- Employee access and privacy policy.
- Operational runbook.
- Named owners for alerts and incidents.

---

## 16. Production Acceptance Criteria

The release is approved for internal production only when:

1. A valid Dynamics event is durably stored before acknowledgment.
2. Duplicate events produce one final analysis.
3. Worker termination does not permanently lose the call.
4. A temporary provider outage recovers automatically.
5. Permanent failures are visible to administrators.
6. Authorized users can view permitted calls.
7. Unauthorized users cannot retrieve calls through either UI or direct API requests.
8. Recordings are encrypted and accessed through controlled links.
9. All schema changes are applied through reviewed migrations.
10. Database backup restoration has been demonstrated.
11. Monitoring detects unavailable workers, queue backlog, database failure, and provider failure.
12. No high- or critical-severity security vulnerabilities remain unresolved.
13. The system passes the defined staging end-to-end scenario.
14. The system passes load testing at twice anticipated peak usage.
15. The company has approved recording, retention, employee-review, and dispute policies.
16. Engineering, operations, security, and the business owner formally approve the release.

---

## 17. Ownership

Suggested ownership model:

- **Product owner:** Sales Operations
- **Technical owner:** Engineering
- **Production operations:** DevOps or Platform Engineering
- **Access management:** IT
- **Security approval:** Information Security
- **Data governance:** Legal, Compliance, or designated privacy owner
- **AI quality owner:** Sales Operations with Engineering support
- **Incident owner:** On-call technical team

Every production alert, integration, data store, and recurring operational task must have a named owner.

---

## 18. Final Product Definition

The internal production release is a secure, single-company sales-call processing and analysis platform that reliably:

- Receives call events from Microsoft Dynamics.
- Processes calls asynchronously.
- Transcribes Persian audio.

- Produces structured AI-assisted sales analysis.
- Stores results durably.
- Provides authorized employees with a searchable dashboard.
- Recovers from temporary failures.
- Exposes failures for operational action.
- Protects recordings, transcripts, employee information, and customer information.
- Maintains sufficient monitoring, auditability, backup, and recovery capabilities for daily internal business use.

The release is production-ready only after the launch blockers and acceptance criteria in this document have been verified through evidence, testing, and formal approval.